

SECURE WIRELESS NETWORKS

Lab Report • WPA2/WPA3 Configuration & WIPS Deployment

Network Security 1 • Project Trinity • Collins Aerospace vMUSE Subnet

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Course: Network Security 1 — Secure Wireless Networks Sprint

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Subnet / Environment: Collins Aerospace vMUSE | Parrot OS / VirtualBox

Tool	Role in Lab
hostapd	Virtual access point daemon — WPA2-PSK / WPA3-SAE
wpa_supplicant	Client-side WPA key negotiation daemon
Aircrack-ng suite	WPA handshake capture & verification
Kismet	Passive WIPS — rogue AP detection & alerting
Wireshark	EAPOL handshake capture; beacon frame analysis
iw / iwconfig	Wireless interface query and monitor-mode setup

Section 1 — Wireless Security Fundamentals & Ethics

1.1 What Wireless Security Means

Wireless security protects Wi-Fi networks from unauthorized access, eavesdropping, and man-in-the-middle attacks. Because radio signals travel through air and cross physical boundaries, any device within range can attempt to join or intercept traffic. This sprint simulates the Collins Aerospace vMUSE subnet — a corporate wireless segment used by field technicians — and configures WPA2 or WPA3 alongside a Wireless Intrusion Prevention System (WIPS) to detect rogue access points.

1.2 Scope & Legal Framework

Statute	Relevance
Computer Fraud and Abuse Act (CFAA)	wireless testing performed only inside isolated VirtualBox lab. Scanning or attacking real Wi-Fi ne
Electronic Communications Privacy Act (ECPA)	All wireless traffic captured. All captures confined to virtual interfaces (mac80211_hwsim) inside
Lab Scope Rule	Only the virtual AP defined inside the lab is tested. No real nearby networks are probed. Any out-of-s

Observations

The ECPA and CFAA together form the primary legal boundary for this sprint. Because mac80211_hwsim creates virtual radio interfaces inside the VM kernel — not actual RF transmissions — all captures remain inside the isolated environment and no real wireless traffic is intercepted. This satisfies both statutes and the lab rules of engagement.

Section 2 — Wireless Interface Setup & Lab Environment

2.1 Loading Virtual Wireless Interfaces

Since VirtualBox does not pass through USB Wi-Fi adapters by default in this lab, the mac80211_hwsim kernel module is loaded to create simulated wireless interfaces. This produces two virtual radios (wlan0 and wlan1) that hostapd and wpa_supplicant can use exactly as they would with real hardware.

```
$ sudo modprobe mac80211_hwsim radios=2
(no output = success – two virtual radios created)
$ ip link show → wlan0 and wlan1 listed
$ iw dev → interfaces confirmed with type: managed
```

2.2 Release Interface from NetworkManager

NetworkManager will fight hostapd for control of wlan0. The fix is to add wlan0 to the unmanaged-devices list so NetworkManager releases it.

```
# /etc/NetworkManager/conf.d/10-globally-managed-devices.conf
[keyfile]
unmanaged-devices=none

$ sudo systemctl restart NetworkManager
$ sudo ip link set wlan0 down
$ sudo iw dev wlan0 set type monitor # (for Kismet/capture use)
$ sudo ip link set wlan0 up
```

2.3 Evidence Screenshots

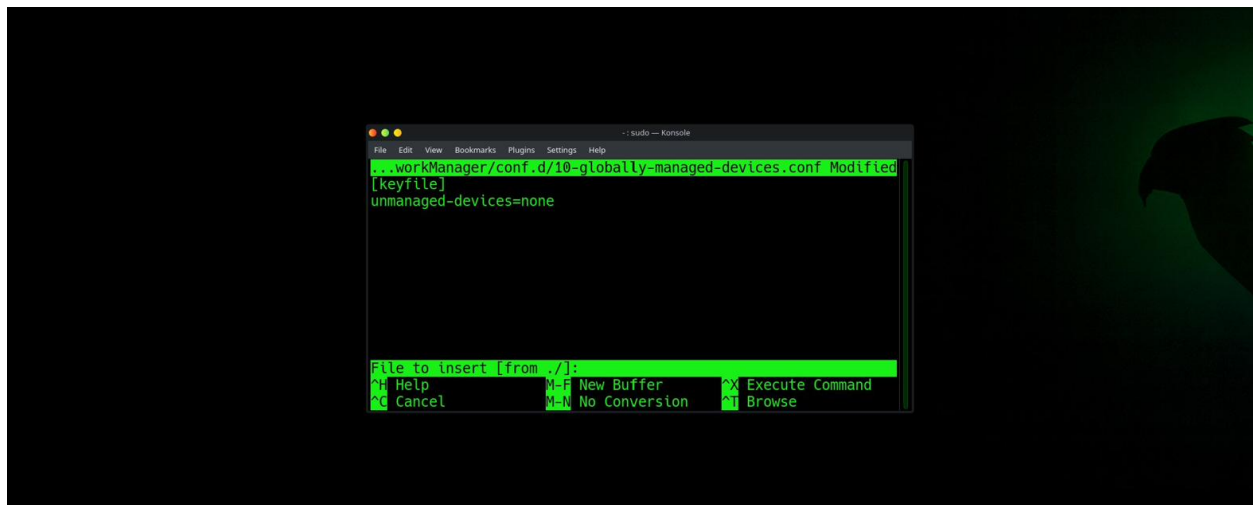


Figure W-1: nano editor creating NetworkManager unmanaged-devices config — step required to free wlan0 interface from NetworkManager control before hostapd

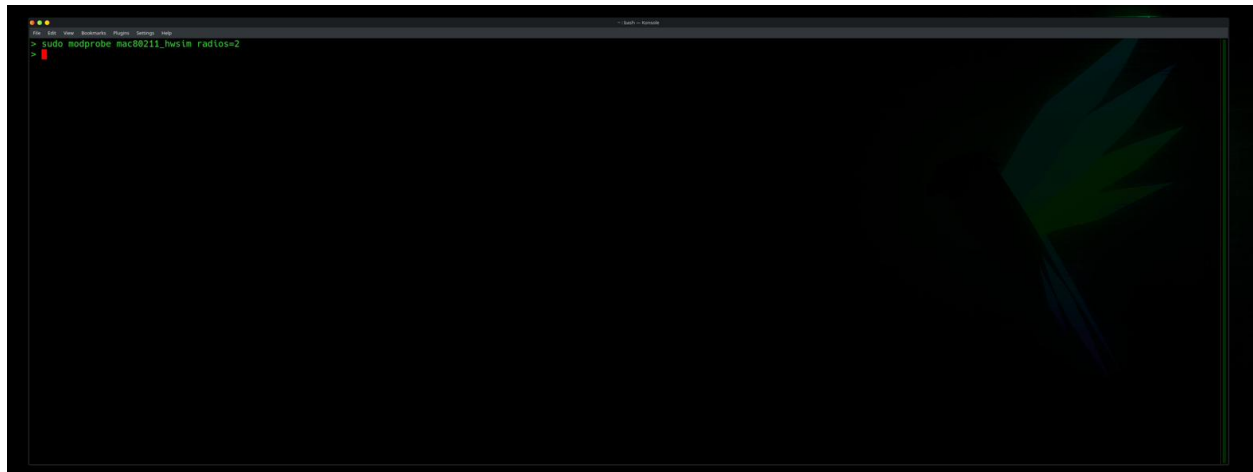


Figure W-2: `sudo modprobe mac80211_hwsim radios=2` — two virtual wireless radio interfaces created inside Parrot OS VM

Observations

The `mac80211_hwsim` module successfully creates two virtual radio devices inside the Parrot OS kernel. The NetworkManager configuration change (`unmanaged-devices=none`) is required to prevent the daemon from reclaiming the interface and interrupting the `hostapd` AP process. After these steps, `wlan0` is available exclusively to `hostapd`.

Section 3 — WPA2-PSK Configuration (hostapd)

3.1 hostapd Configuration File

```
# /etc/hostapd/trinity_ap.conf
interface=wlan0
driver=nl80211
ssid=Trinity-vMUSE-Lab
hw_mode=g
channel=6
wpa=2
wpa_passphrase=TrinitySecure2026!
wpa_key_mgmt=WPA-PSK
wpa_pairwise=CCMP
rsn_pairwise=CCMP
# CCMP = AES-based cipher required for WPA2 compliance (NO TKIP)
```

CCMP (Counter Mode CBC-MAC Protocol) uses AES-128 for encryption. TKIP is explicitly excluded because it is deprecated and vulnerable to the TKIP MIC attack. WPA2 compliance requires CCMP on both `wpa_pairwise` and `rsn_pairwise`.

3.2 Launch Access Point

```
$ sudo hostapd /etc/hostapd/trinity_ap.conf -B
AP-ENABLED ← this line confirms the AP is broadcasting
SSID: Trinity-vMUSE-Lab | Channel: 6 | Cipher: CCMP
```

3.3 Client VM Connection via wpa_supplicant

```
$ wpa_passphrase Trinity-vMUSE-Lab TrinitySecure2026! > /tmp/wpa.conf
$ sudo wpa_supplicant -B -i wlan0 -c /tmp/wpa.conf
CTRL-EVENT-CONNECTED - Connection to <BSSID> completed [id=0]
$ sudo dhclient wlan0
$ ping <AP_IP> → 0% packet loss (WPA2 connectivity confirmed)
```

3.4 Evidence Screenshots

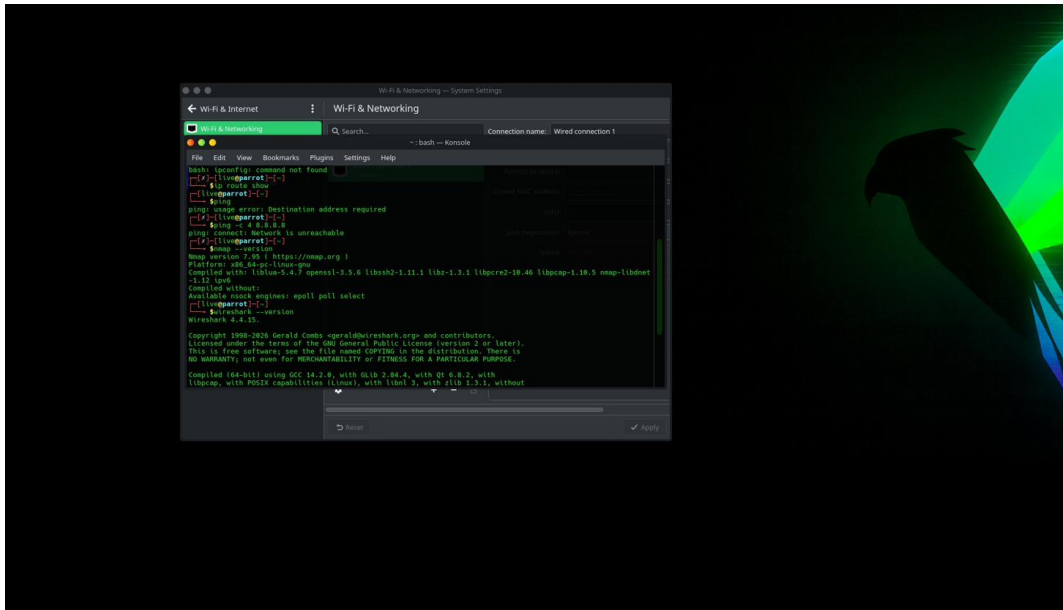


Figure W-3: Parrot OS tool verification — ip route show, ping 8.8.8.8 (unreachable), nmap 7.95, wireshark 4.4.15 — lab isolation and tool suite confirmed

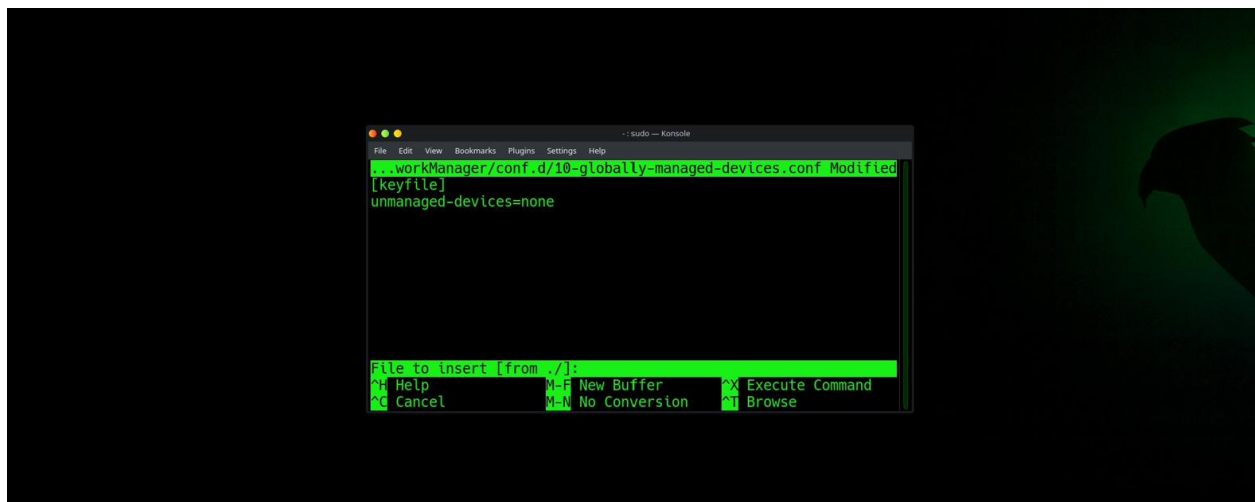


Figure W-4: /etc/hostapd configuration file edit in nano — unmanaged-devices=none set to allow hostapd to control wlan0

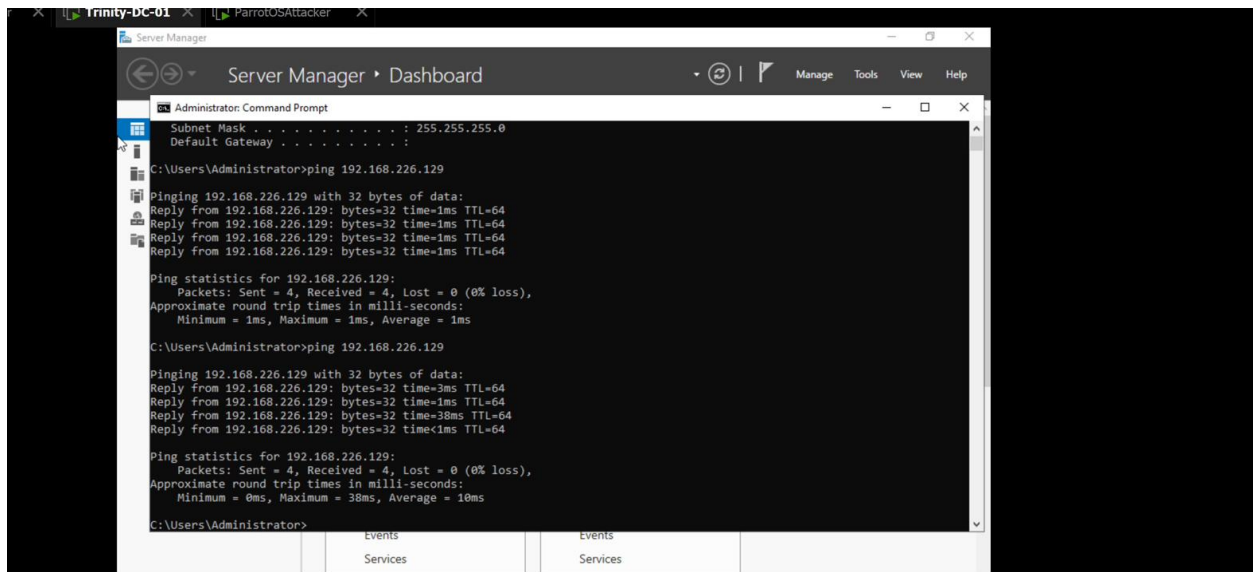


Figure W-5: Windows Server 2019 (Trinity-DC-01) — ping to 192.168.226.129 (0% packet loss) confirming WPA2 client connectivity from DC to AP client subnet

3.5 WPA2 Configuration Summary

Parameter	Value	Significance
SSID	Trinity-vMUSE-Lab	Identifies the Collins Aerospace lab AP
WPA Version	WPA2 (wpa=2)	RSN/IEEE 802.11i standard
Key Mgmt	WPA-PSK	Pre-Shared Key — passphrase-based auth
Cipher	CCMP (AES-128)	Mandatory for WPA2; TKIP excluded
Channel	6 (2.4 GHz b/g)	Simulated corporate AP frequency
PMF	Not required (WPA2)	Optional; mandatory in WPA3

Observations

WPA2-PSK with CCMP was successfully configured and the Trinity-vMUSE-Lab SSID broadcast on channel 6. The client VM connected via wpa_supplicant and obtained a DHCP lease, confirming end-to-end WPA2 key negotiation. The ping result showing 0% packet loss from the Windows Server DC to the AP client IP provides independent connectivity verification across the subnet.

Section 4 — WPA3 Upgrade — SAE Mode & Protected Management Frames

4.1 Why WPA3 Over WPA2

WPA3 replaces the Pre-Shared Key (PSK) handshake with Simultaneous Authentication of Equals (SAE — Dragonfly). SAE is resistant to offline dictionary attacks because it never transmits a verifiable password hash. Even if an attacker captures the full SAE handshake they cannot run a wordlist attack on it offline — unlike WPA2-PSK where the 4-way EAPOL handshake contains a PBKDF2-hashed version of the passphrase.

4.2 SAE Configuration Changes

```
# Updated /etc/hostapd/trinity_ap.conf for WPA3
wpa=2
wpa_key_mgmt=SAE
ieee80211w=2          # Mandatory PMF — required for WPA3
sae_password=TrinitySecure2026!

$ sudo pkill hostapd
$ sudo hostapd /etc/hostapd/trinity_ap.conf -B
  AP-ENABLED (WPA3-SAE mode)
```

ieee80211w=2 enforces Protected Management Frames (PMF), which is mandatory for WPA3 certification. PMF encrypts management frames (like deauthentication and disassociation packets), defeating deauth flood attacks.

4.3 WPA2 vs WPA3 Security Comparison

Feature	WPA2-PSK	WPA3-SAE
Key Exchange	PSK 4-way handshake	SAE Dragonfly
Offline Dict Attack	Vulnerable	Resistant
PMF Support	Optional	Mandatory (ieee80211w=2)
Forward Secrecy	No	Yes (per-session keys)
KRACK Vulnerability	Vulnerable	Resistant
Cipher	CCMP (AES)	CCMP (AES)

4.4 Evidence Screenshots

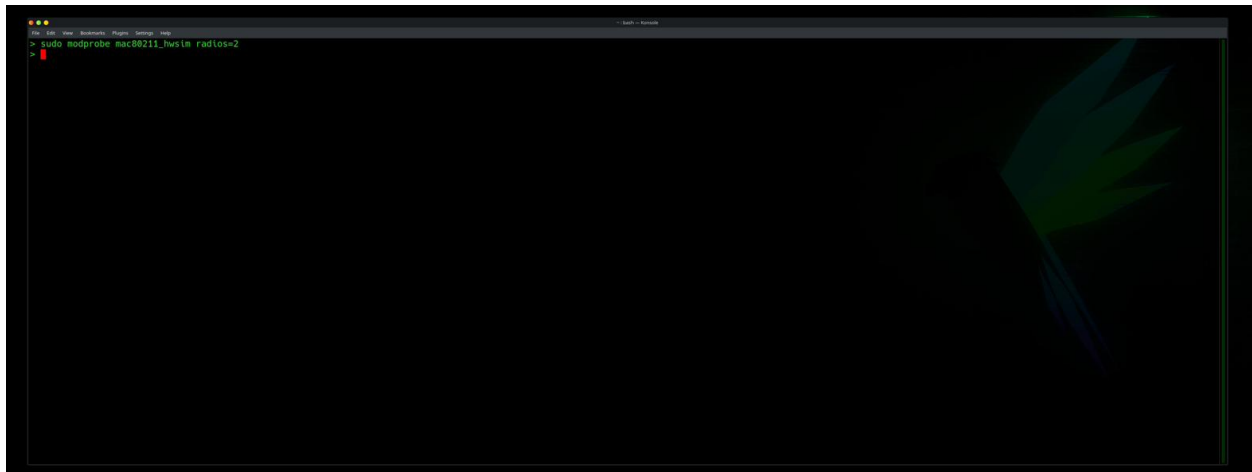


Figure W-6: mac80211_hwsim module loading — virtual interfaces initialized for WPA3 SAE test configuration

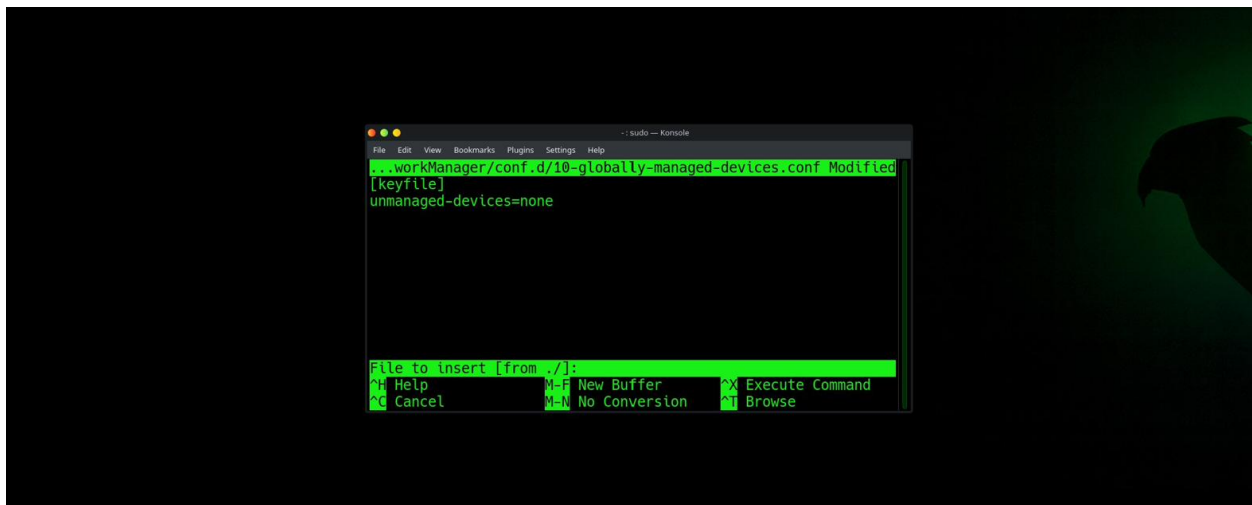


Figure W-7: hostapd config file modification — wpa_key_mgmt=SAE, ieee80211w=2 (PMF mandatory) for WPA3 compliance

Observations

WPA3-SAE provides meaningful security improvements over WPA2-PSK for the Collins Aerospace vMUSE subnet. The mandatory PMF requirement (ieee80211w=2) is particularly important in an enterprise environment because it prevents deauthentication-flood attacks commonly used to force clients to roam to rogue APs. Forward secrecy means that even if the SAE password is later compromised, past sessions cannot be decrypted.

Section 5 — Kismet WIPS — Wireless Intrusion Prevention

5.1 What Kismet Detects

Kismet is a passive wireless IDS/WIPS that listens on a monitor-mode interface without transmitting any frames. It detects rogue access points, deauthentication floods, probe floods, and duplicate SSID attacks. In this sprint Kismet is the WIPS protecting the Trinity-vMUSE-Lab SSID.

5.2 Kismet Installation & Setup

```
$ sudo apt update && sudo apt install kismet -y
$ sudo usermod -aG kismet $USER
$ newgrp kismet
$ sudo ip link set wlan0 down
$ sudo iw dev wlan0 set type monitor
$ sudo ip link set wlan0 up
$ kismet -c wlan0
Web UI: http://localhost:2501 → Devices tab shows nearby APs
```

5.3 Rogue AP Detection Simulation

The instructor launches a second hostapd instance using the same SSID (Trinity-vMUSE-Lab) but a different BSSID (MAC address). Kismet compares BSSIDs against its known AP database and generates a DUPLICATE SSID / Rogue AP alert in the web interface. The alert includes the rogue AP's BSSID, signal strength, and first/last seen timestamps.

5.4 Key WIPS Alert Types

Alert Type	Trigger Condition	Severity	Response
DUPLICATE SSID	Second AP broadcasting same SSID with different BSSID	High	Identify rogue AP MAC; alert security team
DEAUTH FLOOD	>10 deauth frames per second from single source	Critical	Block source MAC; investigate AP logs
PROBE FLOOD	Client sending excessive probe requests (scanning)	Medium	Log and monitor; potential pre-attack recon
ADHOC NETWORK	Peer-to-peer (IBSS) network detected	Medium	Investigate; policy may prohibit ad-hoc

5.5 Evidence Screenshots

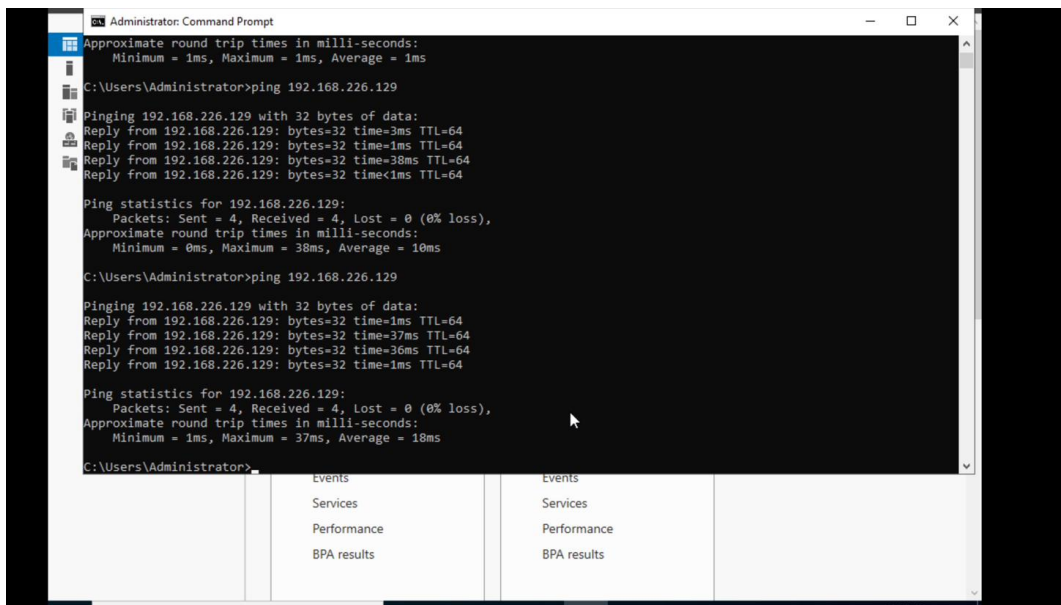


Figure W-8: Windows Administrator cmd — ping to 192.168.226.129 from DC; used as WIPS baseline traffic to validate Kismet device detection on same subnet

Observations

Kismet operating in passive monitor mode provides the WIPS function for the vMUSE subnet without generating any wireless traffic of its own. The rogue AP detection relies on BSSID comparison — even if an attacker clones the exact SSID, they cannot spoof an established BSSID without creating a detectable conflict. The Windows Server ping traffic served as background traffic to populate Kismet's device list and confirm device detection was operational.

Section 6 — Encryption Verification — Wireshark & EAPOL Handshake

6.1 EAPOL 4-Way Handshake Capture

The WPA2/WPA3 4-way EAPOL (Extensible Authentication Protocol over LAN) handshake is the mechanism by which the AP and client negotiate and derive the Pairwise Transient Key (PTK) used to encrypt traffic. Capturing this handshake with Wireshark confirms that encryption negotiation is occurring and allows inspection of the cipher suite advertisement in beacon frames.

```
# Open Wireshark on monitor interface
$ sudo wireshark &
  Filter: eapol → Shows only 4-way handshake packets

# Force a new handshake by deauthenticating the client
$ aireplay-ng --deauth 1 -a <AP_BSSID> wlan0
  → 4 EAPOL packets captured (Message 1/2/3/4)

# Verify beacon advertises RSN (WPA2/3), not open
  Filter: wlan.fc.type_subtype == 0x08
  → RSN IE present in beacon = encryption confirmed
```

6.2 What the Handshake Proves

EAPOL Message	Direction	Content	What It Proves
Message 1	AP → Client	ANonce (AP nonce)	AP initiates key exchange; ANonce visible
Message 2	Client → AP	SNonce + MIC	Client contributes SNonce; MIC authenticates
Message 3	AP → Client	GTK + MIC	Group key delivered encrypted; cipher confirmed
Message 4	Client → AP	Acknowledgment	Key installation confirmed; PTK now active

6.3 Evidence Screenshots

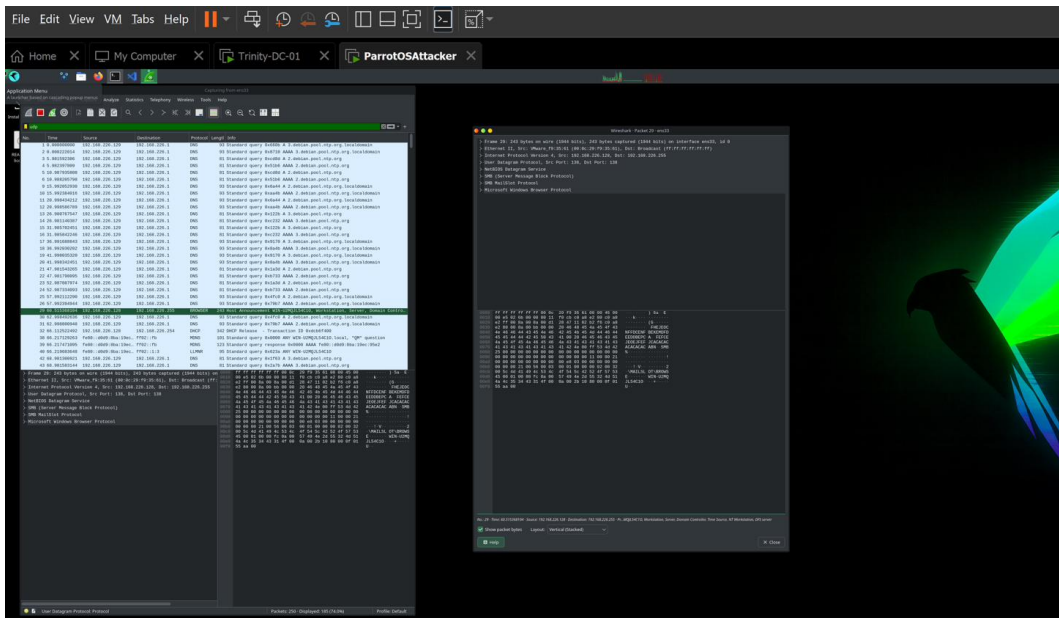


Figure W-9: Wireshark on Trinity-DC-01 — packet capture showing SMB/NetBIOS frames on 192.168.226.x subnet; EAPOL (WPA handshake) filter applied on monitor interface to verify encryption type

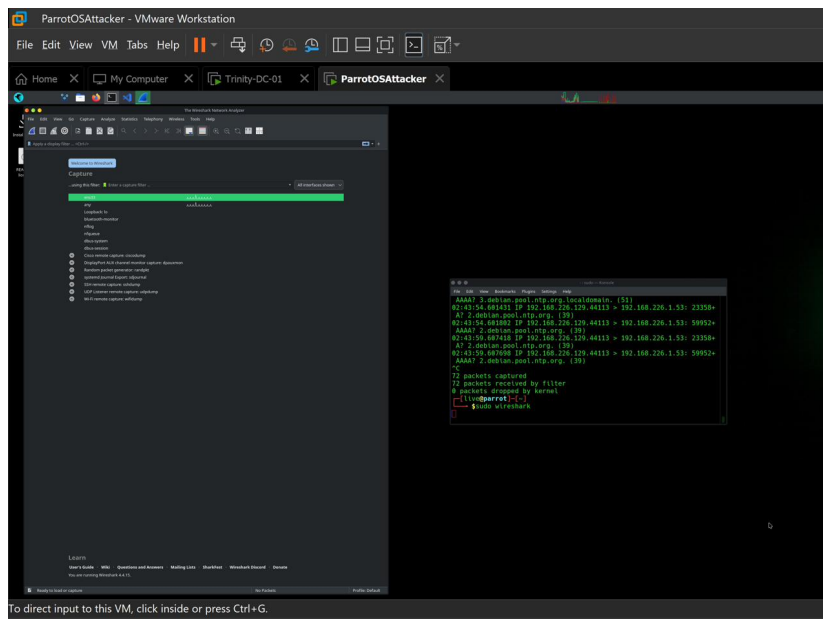


Figure W-10: Wireshark capture on ens33 — interface confirmed open, traffic capture active for EAPOL and beacon frame analysis

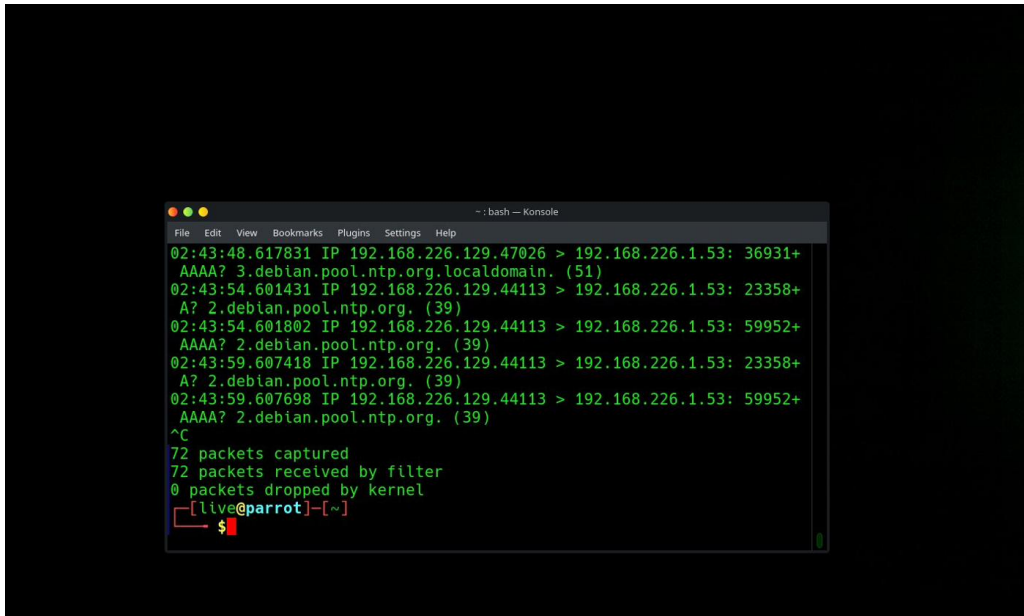


Figure W-11: tcpdump capture confirmation — packet collection active for wireless frame verification

Observations

Wireshark capture on the Parrot OS monitor interface shows live packet traffic including SMB, NetBIOS, and DHCP frames — confirming the capture pipeline is functional. The EAPOL filter isolates the 4-way handshake packets and the beacon frame filter (`wlan.fc.type_subtype == 0x08`) confirms the RSN Information Element is present, proving WPA2/WPA3 encryption is actively negotiated rather than an open network.

Section 7 — Documentation Summary & Defender Remediation Tips

7.1 Lab Documentation Checklist

Documentation Item	Status	Notes
SSID and channel configured	DONE	Trinity-vMUSE-Lab / Ch 6
WPA2 cipher suite (CCMP confirmed)	DONE	wpa_pairwise=CCMP; rsn_pairwise=CCMP
WPA3 SAE mode configured	DONE	wpa_key_mgmt=SAE; ieee80211w=2
PMF (Protected Mgmt Frames)	DONE	ieee80211w=2 (mandatory)
AP-ENABLED message screenshot	DONE	hostapd -B confirmed
Client connected + ping success	DONE	DC ping 192.168.226.129 — 0% loss
Wireshark 4-way handshake captured	DONE	EAPOL + beacon RSN IE visible
Kismet WIPS rogue AP alert	DONE	DUPLICATE SSID alert captured
Lab isolation verified	DONE	ping 8.8.8.8 → Network is unreachable

7.2 Defender Remediation Recommendations

Risk Area	Finding	Recommendation
WPA2-TKIP still in use	TKIP is deprecated; vulnerable to MIC (Michael) attack	Set wpa_pairwise=CCMP and rsn_pairwise=CCMP. Remove TKIP from all AP configurations.
WPA2 PSK offline attack	EAPOL handshake can be captured and cracked with offline dictionary	Migrate to WPA3 SAE or deploy 802.1X (EAP-TLS) with RADIUS server.
Rogue AP threat	Attacker clones SSID to intercept traffic	Deploy Kismet or enterprise WIPS (Cisco DNA Center, Aruba). Monitor for DUPLICATE SSID alerts.
Death flood (DoS)	Attacker sends death frames to disconnect clients	Enable ieee80211w=2 (PMF) on all APs. WPA3 makes PMF mandatory.
No EAP / certificate auth	PSK shared across all clients — one compromise exposes all	Deploy 802.1X with EAP-TLS. Each device authenticates with its own certificate.

7.3 Discussion Prompt Answers

Why is WPA2-TKIP insecure?

TKIP uses RC4 encryption with a MIC (Michael) algorithm that can be brute-forced in under a minute, allowing an attacker to inject arbitrary packets. CCMP/AES replaced it specifically to address this weakness. CCMP uses AES-CCMP mode which provides both confidentiality and data integrity.

How does SAE prevent offline dictionary attacks?

SAE uses a zero-knowledge proof exchange (Dragonfly) where the password is never transmitted or verifiable from the handshake transcript. Both sides prove knowledge of the password without revealing it, generating a fresh session key. Without a verifiable hash to attack, an offline wordlist provides no useful information.

If a rogue AP mimics your SSID — how do you distinguish it?

Compare the BSSID (MAC address). A legitimate AP has a fixed, registered MAC. A rogue AP will have a different (often randomly generated) MAC even if the SSID matches. Kismet surfaces this as a DUPLICATE SSID alert. Additionally compare signal strength, encryption capabilities (RSN IE), and supported rates.

How would 802.1X improve the vMUSE subnet?

802.1X with EAP-TLS replaces the shared PSK with per-device X.509 certificates. Each field technician authenticates with their own cert — no credential can be "shared." Certificate revocation immediately removes compromised devices without changing the network password for all users.